Unit 1 - Equations Only

do they influence the coverage area of AM/FM broadcast

electromagnetic waves travel with a finite speed, equal to the speed of light (($c = 3 \times 10^8\ \text{m/s}$)).

(t_r) = Retarded time

(t) = Present time

(R) = Distance between source and observation point

(c) = Speed of light

They propagate with a finite speed ($c = 3 \times 10^8$) m/s.

(June/Dec University Exam – 7/10 Marks)

A = Magnetic Vector Potential

V = Scalar Potential

($B=\nabla\times A$)

Retarded Potential ($t_r=t-\frac{R}{c}$)

($E_\theta=j60\pi\frac{Idl}{\lambda r}\sin\theta$) Small Dipole

(June 2025 – 7/10 Marks)

$\boxed{L=\frac{\lambda}{4}}$

Length = $\lambda/4$

$\lambda/4$

=

$\boxed{\theta=0^\circ}$

AM/FM Radio Broadcasting

VHF/UHF Communication Systems

(May 2023 – 7/10 Marks)

■= ■0■■■■

■0=Current amplitude

■■=Length of current element

■=Wavelength

■■= ■■0■■

■=Permeability

■=Distance from antenna

■=

■=

■■=

∇× ■= ■■■■

■■= ■

■■= ■■0■■sin ■

■■= 0

■= ■■■■ + ■■■■

■= ■■■■

■■= ■■■■0■■

■■= ■■■0■■

■= 120■Ω

■■= ■60 ■■0■■

■■∝■■■■

■(■) = ■■■■

■= ■■■

■= ■■, ■■■■

■■= ■

■■= Radiated power (W)

■= RMS antenna current (A)

■■= Radiation resistance (Ω)

■■= ■■■

■■= ■■■■(■

■= Length of antenna

■= Wavelength

■=

■■= Radiation resistance

■■= Loss resistance

(Dec 2024 – 5/7 Marks)

■< ■■■

■= Distance from antenna

■= Largest dimension of antenna

Field strength decreases rapidly (■/■■ or ■/■■).

■> ■■■

■= ■■+ ■■+ ■■

■= ■■+ ■■+ ■■

■= ■= ■■■■ ■

■= ■■■■

(June 2025 – 5/7 Marks)

independent of the distance except for the ■/■decrease in amplitude.

■= Maximum dimension of antenna

■= ■■, ■= ■■

■= ■= ■■■■= ■■■ ■

■, ■∝■

Near-field terms (■,/■■ ■/■■)are neglected.

Only the ■/■radiation term is considered.

■= ■× ■

■■= ■■■■■■(■

■■= Transmitted power

■■= Received power

■■= Transmitter antenna gain

■■= Receiver antenna gain

■= Distance between antennas

$P_r=\frac{1}{2}I^2R_r$

(P_r) = Radiated power

(I) = RMS current

(R_r) = Radiation resistance